

NAYAN CHAUDHARY

Software Engineer

Kathmandu, Nepal | +977 9823416640 | nayanchy2004@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Backend Software Engineer with hands-on experience designing reliable, secure, and scalable applications using Java and Spring Boot. Experienced in building transaction-oriented systems, RESTful services, and real-time platforms with a strong focus on data consistency, authentication, asynchronous processing, and performance optimization. Passionate about solving complex backend problems through clean architecture, robust system design, and maintainable code.

EXPERIENCE

Web Development Intern — Nepal Electricity Authority

Sept 2025 – Dec 2025

- Designed and improved **responsive UI components** for internal government web applications using React.js.
- Contributed to enhancing user experience by refining layouts, accessibility, and front-end interactions.
- Proposed and designed a **dictionary-based Nepali-to-English text conversion logic**, improving bilingual usability of the system.
- Collaborated with backend developers to align UI with REST API data flows and application logic.
- Gained hands-on experience in **full-stack development practices**, focusing primarily on frontend implementation.

PROJECTS

Fintech Wallet & Payment Processing System :- [Github](#)

Java, Spring Boot, Spring Security, JWT, MySQL, Redis, Apache Kafka, Docker, JPA/Hibernate, JUnit, Test containers

- Built a **scalable backend system** for digital wallet operations including P2P transfers, deposits, and withdrawals via 20+ secured REST APIs.
- Designed a **double-entry ledger system** ensuring transaction consistency and full auditability.
- Implemented **concurrency-safe transactions** using pessimistic locking to prevent race conditions under concurrent load.
- Developed **idempotent payment processing** with Redis-based keys to eliminate duplicate transactions.
- Integrated **Apache Kafka for asynchronous processing**, with retry and dead-letter queue (DLQ) handling for failure resilience.
- Validated production readiness with **160+ JUnit/Testcontainers unit and integration tests** (MySQL & Kafka) covering 20+ REST endpoints across auth, wallet, transaction, and notification modules; documented the full API contract via Swagger/OpenAPI.

Community EV Station Platform:- [Backend](#) | [Frontend](#)

Spring Boot, Spring Security, JWT, MySQL, JPA, WebSocket, STOMP, Spring Cache, Spring Scheduler, React.js

- Built a full-stack Airbnb-style marketplace enabling residents to list private EV chargers and drivers to book guaranteed charging slots in advance.
- Designed 30+ REST endpoints across Admin, Host, and User roles covering listings, bookings, payments, analytics, real-time chat, and notifications.
- Secured the application with **JWT authentication and role-based access control (RBAC)**.
- Automated booking lifecycle and system events using **Spring Scheduler and caching mechanisms**.
- Built **real-time chat system** using WebSocket (STOMP/SockJS) with offline queueing and auto-reconnect.

SKILLS

Languages: Java, JavaScript, SQL, C, C++

Backend: Spring Boot, Spring Security, Spring Data JPA/Hibernate, JWT & RBAC, WebSocket (STOMP), Apache Kafka, REST API Design, Maven

Infrastructure & Data: MySQL, Redis, Docker & Docker Compose, Lua Scripting, Idempotency & Rate Limiting

Tools & Practices: JUnit, Test containers, Postman, Swagger/OpenAPI, Git, GitHub, IntelliJ IDEA, System Design, Concurrency Control

EDUCATION

Bachelor of Engineering in Information Technology

2022 – 2026

Cosmos College of Management and Technology, Kathmandu, Nepal

LANGUAGES

Nepali (Native) | English (Professional)